

A.4 Combinatorics

1. Permutations

- complete lists $\rightarrow$ how many possible rankings are there for a grp. of 3 people.

$$\underline{3} \times \underline{2} \times \underline{1} = 6$$

- how many for 12 people?

$$12 \times 11 \times 10 \dots \times 3 \times 2 \times 1$$

- Idea $\rightarrow$ At each step, multiply the no. of choices with the next step with choices so far.

$$n! = n \times (n-1) \times (n-2) \times \dots \times 2 \times 1$$

$$1! = 1$$

$$0! = 1$$

- Each order is cld a permutation, $n!$ is the no. of permutations of n objects.
- Partial lists $\rightarrow$ a contest of 12 people with 3 distinct prizes. How many ways?

$$\underline{12} \times \underline{11} \times \underline{10} \rightarrow \text{eg. of partial or } k \text{ permutation.}$$

eg. How many 3 digit no. are with all even digits?

$$\underline{4} \times \underline{5} \times \underline{5} = 100$$

Eg. same ques + no. repeated digits

$$\underline{4 \times 4 \times 3} = 48 \rightarrow \text{this time we choose digits w/o replacement.}$$

2. Combinations

Eg. 12 people, 3 prizes, what if 3 prizes were same.

$${}^{12}C_3 = \frac{12 \times 11 \times 10}{3 \times 2 \times 1}$$

- To choose r items from a group of n , if all items we choose are equal then the no. of ways to do so -

$${}^nC_r = \frac{n!}{(n-r)! r!}$$

$$\binom{n}{r} = \binom{n}{n-r}$$

$${}^nC_0 = {}^nC_n = \frac{n!}{0! \cdot n!} = 1$$

Eg. 18 people and 3 groups. $\rightarrow 8, 6, 4$. How many divs. are possible?

$$\text{Pick up group of 8} \Rightarrow \frac{18!}{8! \cdot 10!}$$

$$\frac{18!}{8! \cancel{10!}} \times \frac{\cancel{10!}}{6! \cancel{4!}} \times \frac{\cancel{4!}}{\cancel{4!}} = \frac{18!}{8! 6! 4!}$$

Ex. 4 nos. are picked up from $\{1 \dots 30\}$. How many ways possible s.t. all are divisible by 3.

$${}^{10}C_4 = \frac{10!}{6! 4!} = \frac{10 \times \cancel{9} \times \cancel{8} \times 7}{\cancel{4} \times \cancel{3} \times \cancel{2} \times 1} = 210$$

Ex. s.t. 3 are divisible by 5 & other by 7.

$\begin{matrix} \swarrow & \searrow \\ 6 & 4 \\ \{5, 10, 25, 20, 25, 30\} & \{7, 14, 21, 28\} \end{matrix}$

$$\cancel{6}C_3 \cdot 4C_1 = \frac{6!}{3! 3!} = 20, \quad \frac{4!}{3! \times 1} = 4$$

$$20 \times 4 = 80$$

3. Binomial Distribution

Ex. str shots at 0.7, each shot independent.

(i) probab. (next 4 shots & miss 2)

(ii) probab. (exactly 4 shots out of 6)

(i) $P(\text{shot})^4 P(\text{no shot})^2 = (0.7)^4 (0.3)^2$

(ii). ${}^6C_4 \times (0.7)^4 \times (0.3)^2$

- _/_/_
- A bernoulli (p) random variable (0-1).
A case with $X=1$ is success & $X=0$ is failure.

$$E[X] = p$$

$$\text{Var}(X) = p(1-p)$$

- X is a binomial (n, p) RV if X is the no. of successes in n independent trials, each with a success of same probab. p .

$$P(X=k) = {}^nC_k \cdot p^k \cdot (1-p)^{n-k}$$

→ fixed no. of trials

→ trials are independent

→ success probab. is same for all trials.

$$E[X] = np$$

$$\text{Var}(X) = np(1-p)$$

eg. $N \sim \text{Binomial}(32, 0.9)$

$$\begin{aligned} P(N > 30) &= P(N=31) + P(N=32) \\ &= {}^{32}C_{32} \cdot (0.9)^{32} + {}^{32}C_{31} \cdot (0.9)^{31} \cdot (0.1)^1 \\ &= 0.156 \end{aligned}$$

$$\begin{aligned} E[X] &= 32 \times 0.9 \\ &= 28.8 \end{aligned}$$

$$\begin{aligned} \text{Var}(X) &= 28.8(0.1) \\ &= 2.88 \end{aligned}$$

4. Multinomial Distributions

Ex. accidents - 0.5 (minor), 0.4 (moderate) & 0.1 (severe). 2 accidents in 1 month. Find the probab. that neither is severe & at most one is moderate.

$$\begin{aligned} P(\text{"}) &= P(2 \text{ minor}) + P(1 \text{ minor}, 1 \text{ moderate}) \\ &= {}^2C_2 (0.5)^2 + {}^2C_1 (0.5)(0.4) \\ &= 0.65 \end{aligned}$$

- n trials with 3 possible outcomes

$$P(X_1 = k_1, X_2 = k_2, X_3 = k_3) = \frac{n!}{k_1! k_2! k_3!} p_1^{k_1} p_2^{k_2} p_3^{k_3}$$

→ this can be generalized to r possible outcomes. This is same as binomial distribution. Only that we have more than 2 outcomes.

5. Hypergeometric Distribution

- Non-independent draws → out of a crate of 10 4 are defective. If 3 are randomly selected, what is probab. that at most 1 is defective.

$$P(\text{at most 1 def.}) = P(\text{no def.}) + P(2 \text{ def, 1 no})$$

$$\left. \begin{aligned} P(\text{no def.}) &= \frac{{}^6C_3}{{}^{10}C_3} \\ P(2 \text{ def, 1 no}) &= \frac{{}^6C_2 {}^4C_1}{{}^{10}C_3} \end{aligned} \right\} + = \frac{2}{3}$$

- We have G good pieces out of N total.

$$\text{probab. of choose } g \text{ good} = \frac{{}^G C_g {}^{N-G} C_{n-g}}{{}^N C_n}$$

→ this is not a binomial distribution, know whether or not the first item is good gives info about whether or not the second one will be good

- A binomial needs each trial to be independent. This occurs when the sampling is with replacement. In the hypergeometric distribution we sample w/o replacement in a finite population.

Ex. I have 16 B & 4 W socks. I pick 6 for trip. $P(4 \text{ B \& } 2 \text{ W})$

$$= \frac{{}^{16}C_4 \times {}^4C_2}{{}^{20}C_6} = 0.282$$

Ex. 10 B, 6 Br & 4 W socks. I take 6.
 $P(2 \text{ of each colour})$

$$\frac{{}^{10}C_2 \cdot {}^6C_2 \cdot {}^4C_2}{{}^{20}C_6} = 0.104$$

- If I do things independently, that becomes the binomial distribuⁿ.